

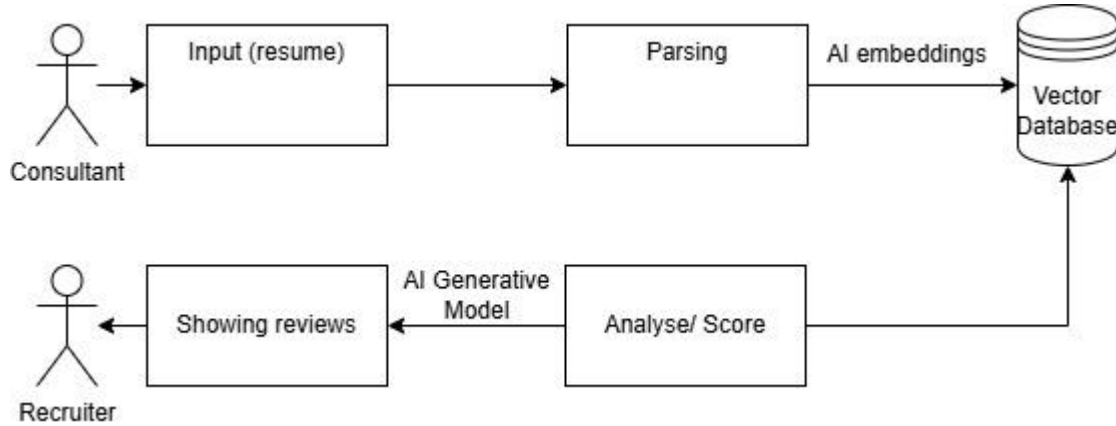


ConsultantAI Match

A proof of Concept

November 2025

Architecture - Draft



Some experiments with Parsing component

- Regular expression → **didn't match the varied form**
- Natural Language Processing (NLP)
 - *resume_parser* library
 -  Match with varied form
 -  Less specific key words → it doesn't be trained well
 - Gemini/ ChatGPT (Open AI API) → paid by token
 -  Match correctly and more specific key words → depends on model

Results of Parsing component

- Gemini

```
{
  "Name": "Quang-Huy Nguyen",
  "Email": "salj@adabeat.com",
  "Phone": null,
  "Work Experience": [
    {
      "Title": "System Designer / Linux Network Engineer",
      "Company": "SkyLab",
      "Dates": "2023-05 - 2025-07",
      "Description": [
        "Engineered a core OSS/BSS (Operational Support System) for a telecommunications provider.",
        "Designed and implemented concurrency and distributed systems.",
        "Implemented a multi-cloud Cloud Abstraction Layer (CAL) for multi-tenant cloud services.",
        "Integrated gRPC/protobuf-based interfaces for microservices.",
        "Deployed microservices to Kubernetes.",
        "Authored technical handover documentation for system transitions.",
        "Collaborated with international product and engineering teams."
      ],
      "TechStack": [
        "Golang",
        "MongoDB",
        "Redis",
        "Kafka messages",
        "MySQL",
        "Kubernetes"
      ]
    }
  ],
}
```

- ChatGPT (open AI)

```
{
  "name": "Quang-Huy Nguyen",
  "email": "salj@adabeat.com",
  "phone": null,
  "work_experience": [
    {
      "title": "System Designer / Linux Network Engineer",
      "company": "SkyLab",
      "period": "2023-05 - 2025-07",
      "description": "Engineered OSS/BSS bill of material system for telecom providers",
      "tech_stack": ["Golang", "MongoDB", "ReactJS"]
    },
    {
      "title": "Associate Research Engineer",
      "company": "Namboo Solution / Korea Railroad Research Institute",
      "period": "2022-01 - 2022-12",
      "description": "Improved odometry accuracy by 10% for autonomous train navigation",
      "tech_stack": ["NodeJS", "VueJS", "NetWorkX"]
    },
    {
      "title": "Research Assistant",
      "company": "Distributed Computing Lab, KAIST",
      "period": "2019-09 - 2021-12",
      "description": "Led benchmarking research on distributed computing frameworks",
      "tech_stack": ["Research", "Kubernetes", "GoLang"]
    }
  ]
}
```

gemini-2.5-flash

```

    "Name": "CCNA course",
    "Issuer": "BKACAD, Hanoi University of Technology",
    "Year": null
  },
  {
    "Available Time": null
  }
]

```

GPT 5

```
"certifications": [
  "Web Development w/ Google's Go (Golang)",
  "Agile Scrum from A to Z - Udemy, 2025",
  "CCNA course - BKACAD, Hanoi University",
],
"available_time": "Available after July"
```

Updating status

- Reading docx files: `CV.docx` and `assignment.docx`
- Calling ***gemini-2.5-flash*** model API to parse CVs and assignments
 - CV: Name, Email, Phone, Work Experience, Education, Skills, Certifications, and Available Time
 - Assignment: Start date, Description, Duration, Location, Working mode
- Storing Consultants' information and Assignments into **MongoDB** database
- Embedding Work Experience and Description by ***gemini-embedding-001*** *EmbedContentConfig(task_type="SEMANTIC_SIMILARITY")* and storing into the database
 - <https://ai.google.dev/gemini-api/docs/embeddings>

Mongo database

The screenshot shows the MongoDB Atlas web interface. The breadcrumb navigation is `consultantatamatch.ftccqos.mongodb.net > project > consultants`. The 'Documents' tab is selected, showing 1 document. The search bar contains the text 'Type a query: { field: 'value' } or [Generate query](#)'. The document being viewed is a JSON object representing a consultant:

```
{
  "_id": ObjectId('690988e605901702d0a21fed'),
  "Name": "Quang-Huy Nguyen",
  "Email": null,
  "Phone": null,
  "Work Experience": Array (4)
    0: Object
      title: "System Designer / Linux Network Developer"
      company: "SkyLab"
      dates: "2023-05 - 2025-07"
      description: Array (7)
        0: "Engineered a core OSS/BSS (Operational Support/ Business Support System)"
        1: "Designed and implemented concurrency processing using goroutines to ef..."
        2: "Implemented a multi-cloud Cloud Abstraction Layer to unify resource ma..."
        3: "Integrated gRPC/protobuf-based interfaces and gRPC-Gateway for REST co..."
        4: "Deployed microservices to Kubernetes; used Kafka, Redis, MongoDB, and ..."
        5: "Authored technical handover documentation and runbooks so on-call/supp..."
        6: "Collaborated with international product and engineering teams, partici..."
      tech_stack: Array (6)
        1: Object
        2: Object
        3: Object
    Education: Array (2)
    Skills: Array (22)
    Certifications: Array (3)
    Available Time: null
    work_experience_embedding: Binary.createFromBase64('YoVtPLLLgDyt3ds8rgmQvShh8jtR/Fm8GhsG07hXr7wVwL8YzC20x8IN7wqYy...
```

The screenshot shows the MongoDB Atlas web interface. The breadcrumb navigation is `consultantatamatch.ftccqos.mongodb.net > project > assignments`. The 'Documents' tab is selected, showing 0 documents. The search bar contains the text 'Type a query: { field: 'value' } or [Generate query](#)'. The document being viewed is a JSON object representing an assignment:

```
{
  "_id": ObjectId('69099da95997277a8a7deb4e'),
  "Start date": "2025-11-03",
  "Description": "We're looking for a UX Designer who is passionate about creating intuiti...",
  "Duration": "Approximately 5 months (from 2025-11-03 to 2026-03-31), Full-time (100%)",
  "Location": "Stockholm",
  "Working mode": "On-site (inferred from location)",
  "description_embedding": Binary.createFromBase64('uTnWPNDprTtqIrs773CPveoYxzstRsU7wbnuv5ZgCrqogQs8qR2e0...')
}
```

Comparison result

- Comparing Experience with Assignment description
 - **1.0** → identical
 - **0.7+** → very similar
 - **~0.0** → unrelated
 - **negative** → opposite meaning

```
(venv) adabeat@LAPTOP-CTY:~/workspace/AImatch$ python comparing_vectors.py
Cosine similarity between work experience and assignment description [69099da95997277a8a7deb4e]: 0.5453
Cosine similarity between work experience and assignment description [69099ecd68b2545345d9cecc]: 0.6058
Cosine similarity between work experience and assignment description [690b098fdb3291f5e2459f02]: 0.5949
Cosine similarity between work experience and assignment description [690b09e669f4a6e5cb94b1dc]: 0.5484
Cosine similarity between work experience and assignment description [690b0f98e940c8bd787ec9ea]: 0.6833
(venv) adabeat@LAPTOP-CTY:~/workspace/AImatch$
```

Questions

1. Many cases are using AI for semantic search → What AI platforms will we rely on?
 - Reliability → accuracy, good AI model
 - Availability → no downtime
 - Confidentiality → don't store personal information
2. What does mathematics use for comparison to have 'good' result?
 - Depend on the embedding models

Thank you



